Genome Comparison using FastGA/FasTan + OneFiles

GENE MYERS, Richard Durbin
rd109@cam.ac.uk

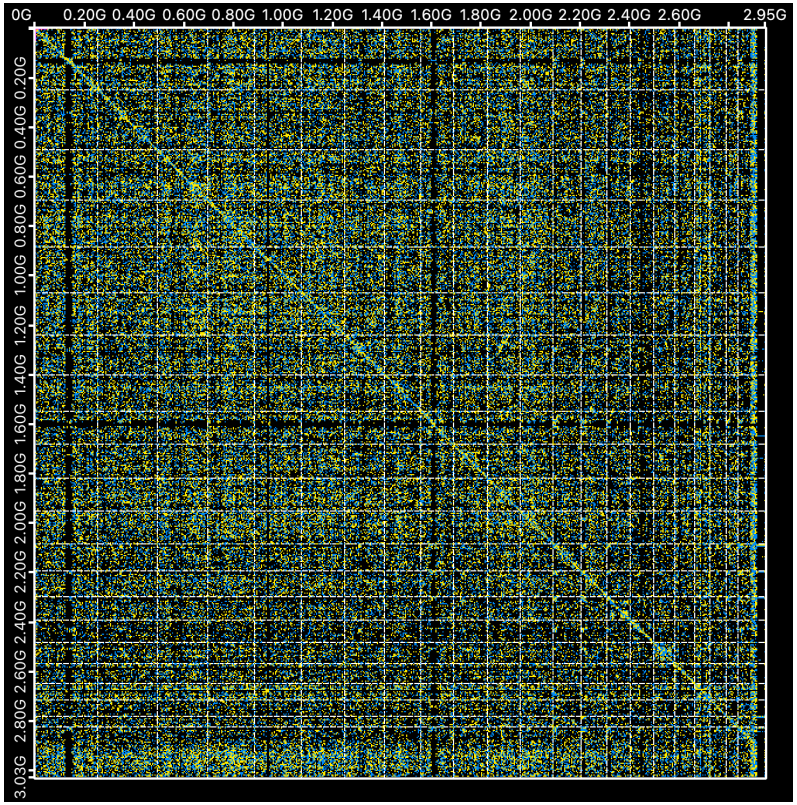
15 October 2026

Scalable Genomics and Pangenomics 2026

Genome Comparison

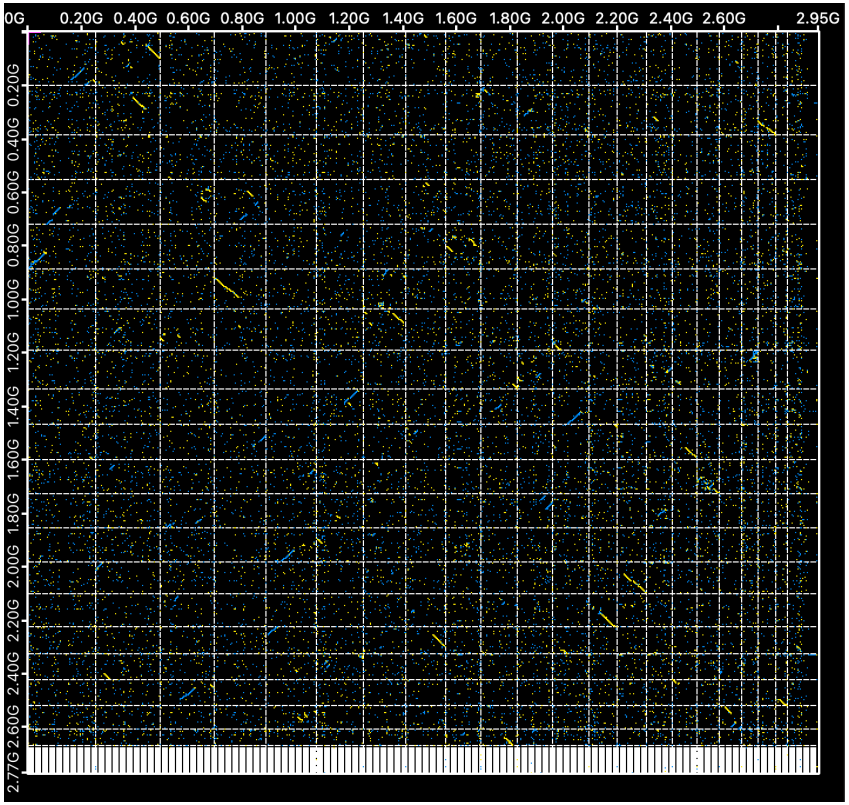
- Based on (k-mer) content:
  - sketching
  - mumemto
  - syng pangenomes – (coloured) de Bruijn graphs
- Based on alignment:
  - MUMmer/NUCmer – early fast use of MUMs as alignment seeds
  - LASTZ – sensitive but slow
  - minimap, …
  - (minigraph)cactus, pggb pangenomes

Genomes differ by mutation since their common ancestor - mostly order and orientation preserving ⇒ colinearity



~7Myr
~1.2% neutral divergence

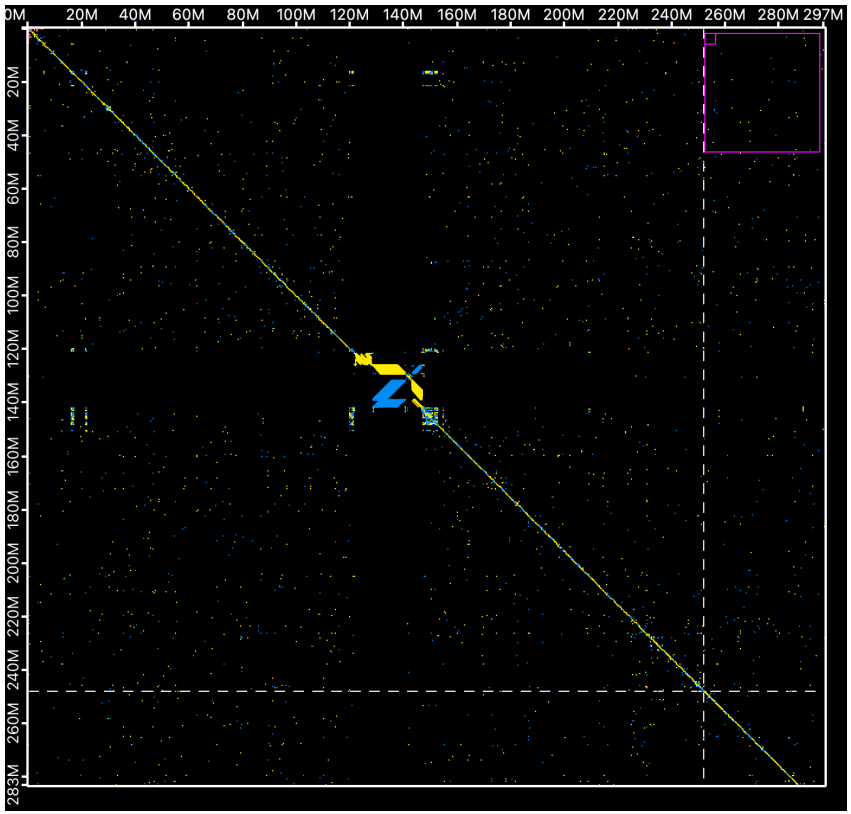
Human – Chimp Whole Genome



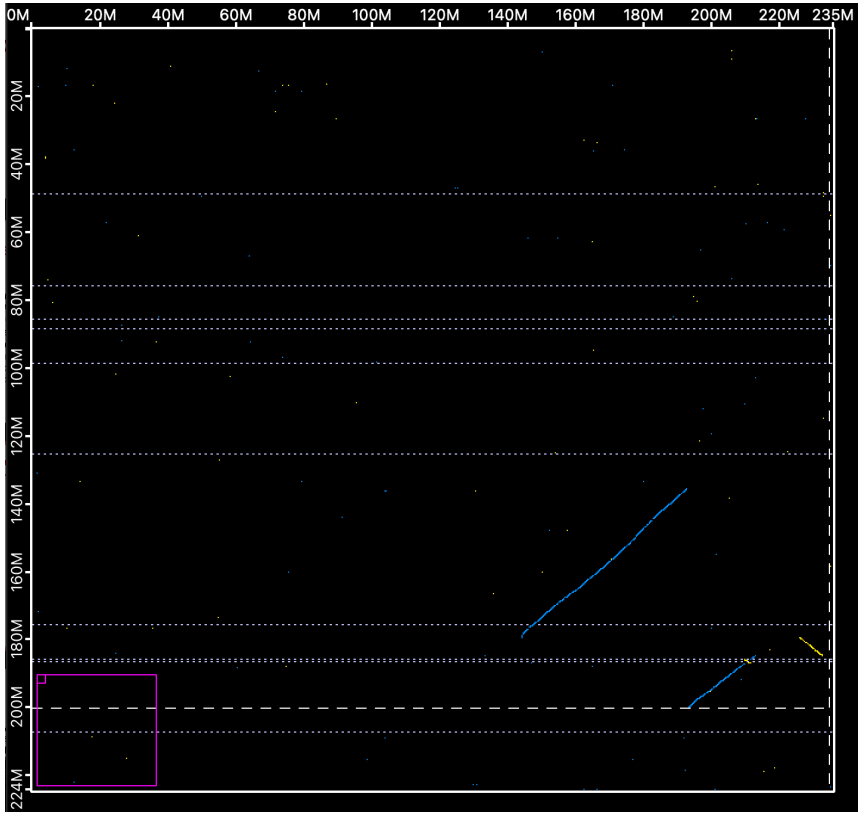
Human – Mouse Whole Genome

~70Myr
~65% neutral

Genomes differ by mutation since their common ancestor - mostly order and orientation preserving ⇒ colinearity

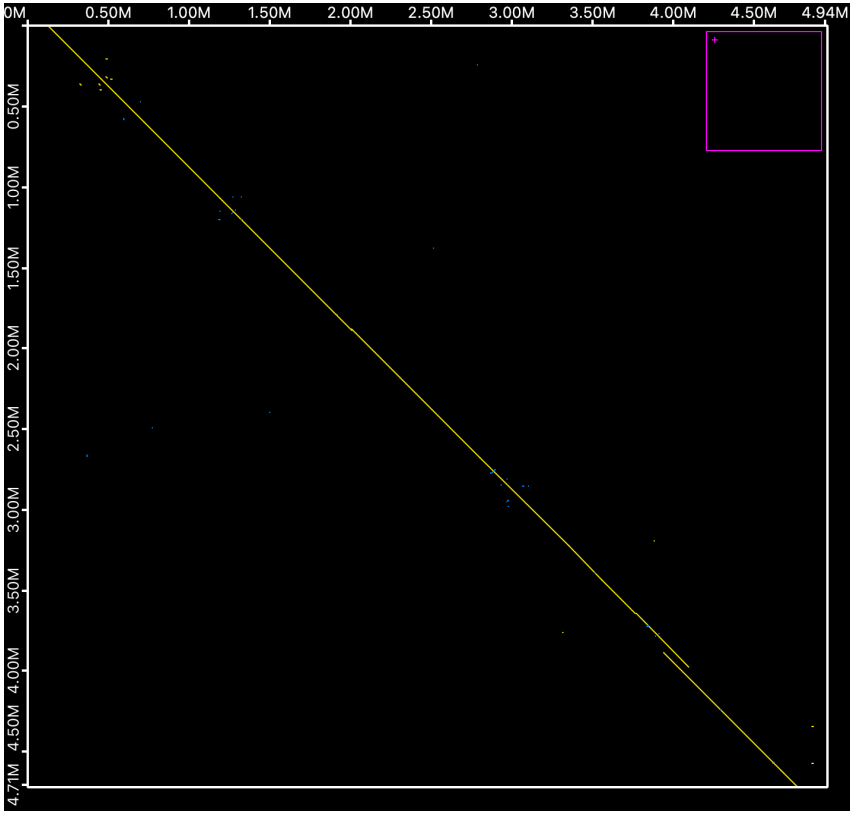


Human – Chimp Chromosome 1 **~10%**

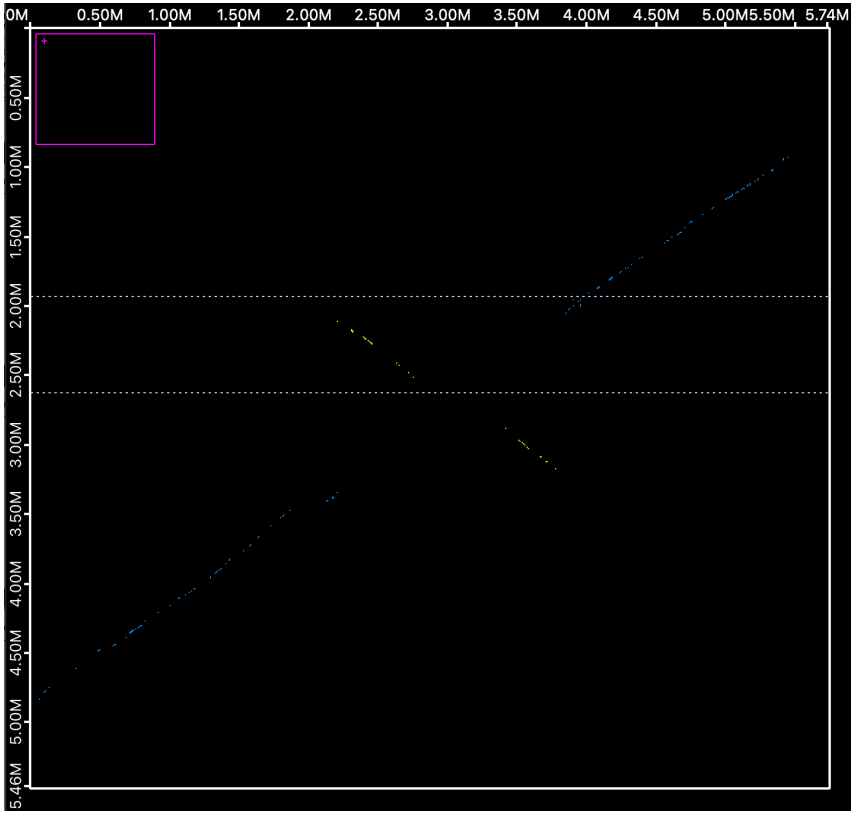


Human – Mouse Chromosome 1 **~10%**

Genomes differ by mutation since their common ancestor - mostly order and orientation preserving ⇒ colinearity

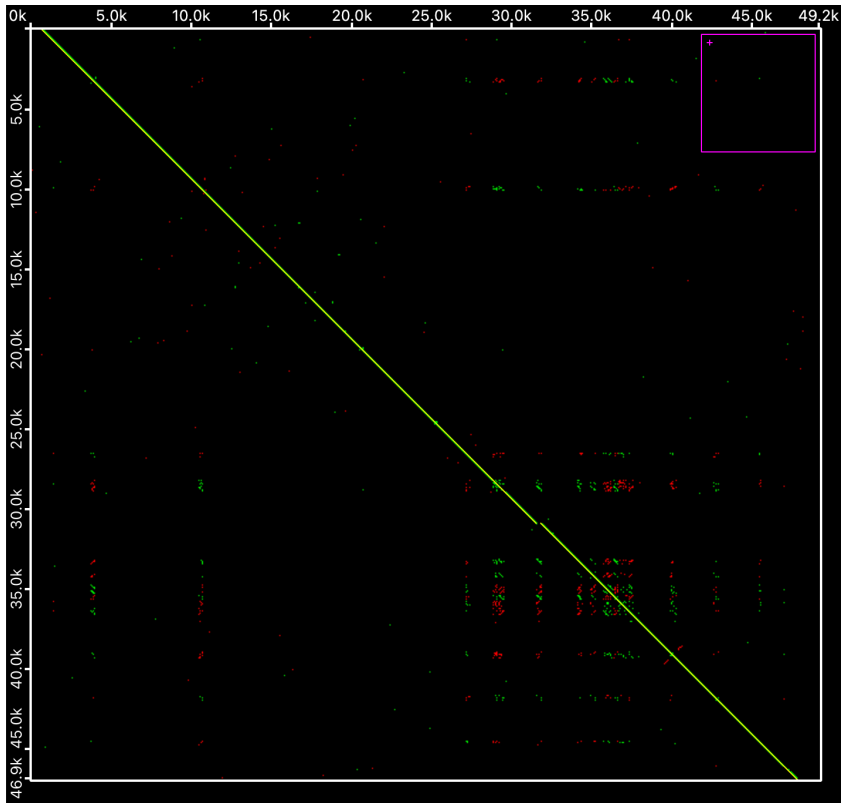


Human – Chimp ~5Mb **~2%**

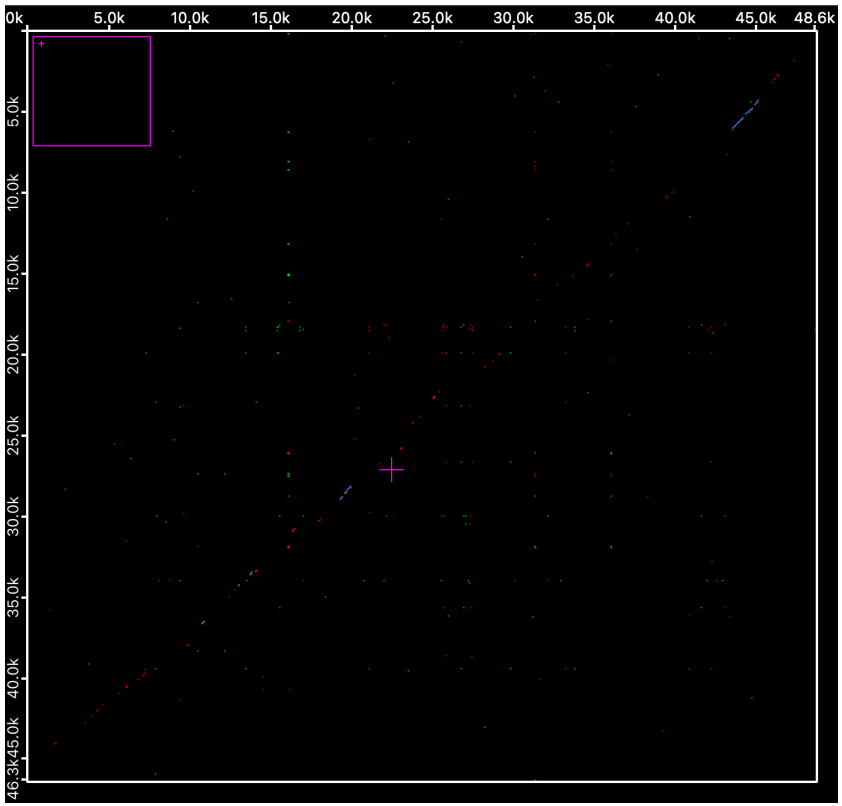


Human – Mouse ~5Mb **~2%**

Genomes differ by mutation since their common ancestor - mostly order and orientation preserving ⇒ colinearity



Human – Chimp ~50kb **~1%**



Human – Mouse ~50kb **~1%**

Essentially all aligners find short exact ***seed*** matches, then **join/extend** them using local dynamic programming

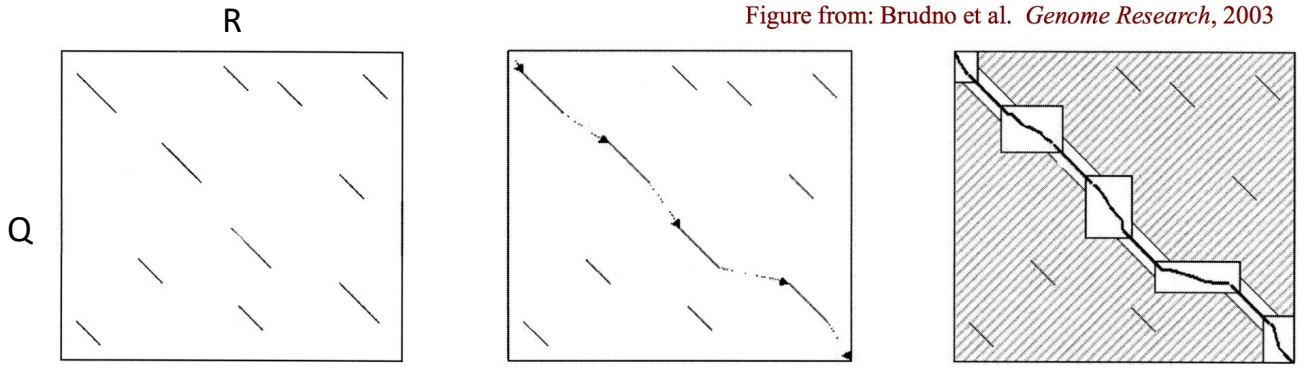


Figure from: Brudno et al. *Genome Research*, 2003

1. perform pattern matching to find seeds for global alignment
2. find a good chain of anchors
3. fill in remainder with standard but constrained alignment method

A seed match is a short diagonal length L in an RxQ search matrix

Each dynamic programming box is small, so (relatively) fast

Design choices

- How to find the seeds:
  - MUMs via suffix trees – `nucmer`
  - MEMs via BWT/FM-index – `bwa-mem, bowtie`
  - minimiser hash – `minimap`, …
  - finite state machine - `blast`
- How to chain: a *dark art*
- How to allow for score the alignment – particularly gaps
  - Linear: cost per gap character - `nucmer`
    - simplest is Levenshtein -1 for a mismatch or gap, apply a percent identity threshold
  - Affine: higher cost to start a gap, then lower cost to extend – `lastz`
  - Multi-affine: make long gaps even cheaper - `minimap`

Design choices

FastGA

- How to find the seeds:
  - MUMs via suffix trees – nucmer
  - MEMs via BWT/FM-index – bwa-mem, bowtie
  - minimiser hash – minimap, …
  - finite state machine - blast

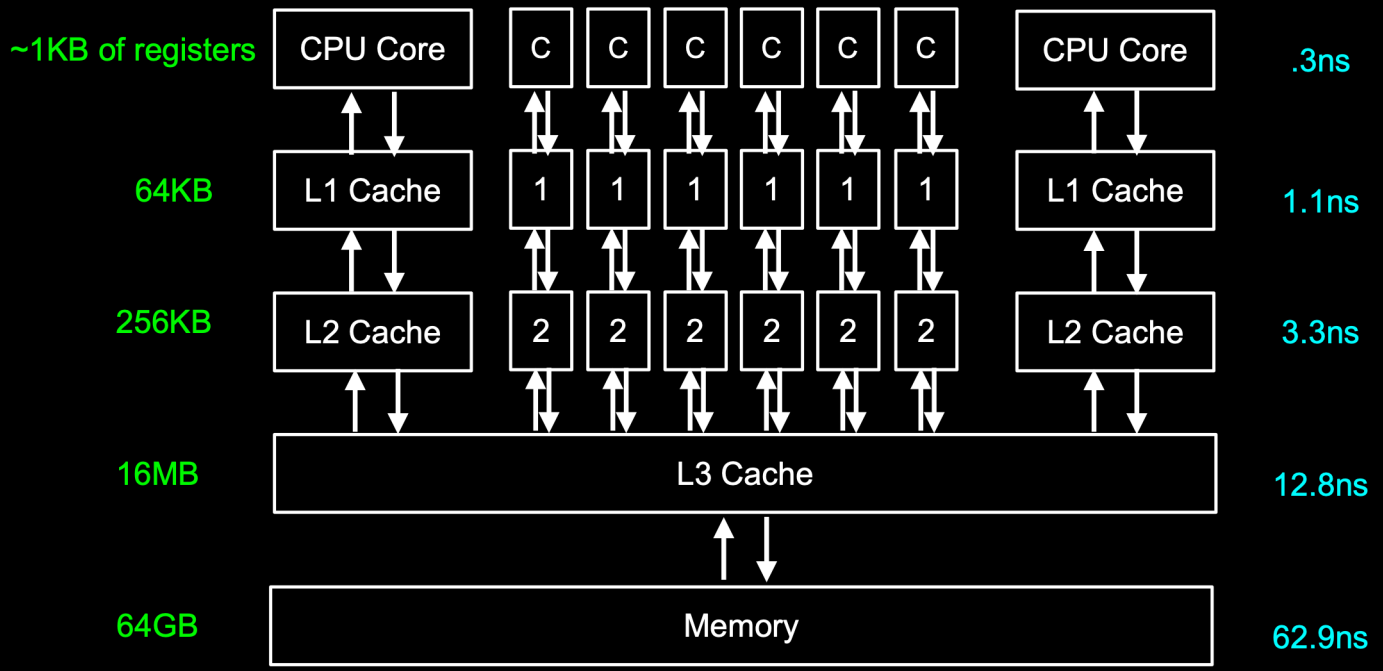
Semi-MEMs via sorting

- How to chain: a *dark art*
- How to allow for score the alignment – particularly gaps
  - Linear: cost per gap character - nucmer
    - simplest is Levenshtein -1 for a mismatch or gap, apply a percent identity threshold

Levenshtein 70%

  - Affine: higher cost to start a gap, then lower cost to extend – lastz
  - Multi-affine: make long gaps even cheaper - minimap

A Cache Miss Can Cost You 60-100 CPU Cycles



Gene Myers

SORTING FOR SPEED

Want "seed" k-mer hits between G_1 & G_2

(*pairs of positions (p_1,p_2) where G_1 & G_2 have the same k-mer*)

Build an index S of k-mers in G_2

For each position p_1 of G_1
Find positions p_2 of k-mer at p_1 in S

not cache coherent

Build list of k-mer,position pairs of G_1 and sort on k-mer

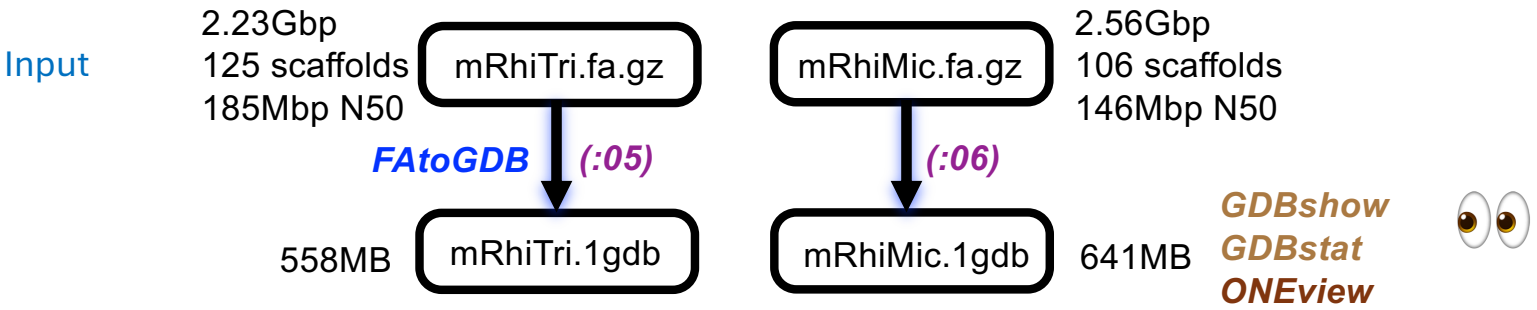
Build list of k-mer,position pairs of G_2 and sort on k-mer

Merge G_1 & G_2 lists by k-mer and build a list of (p_1,p_2) pairs

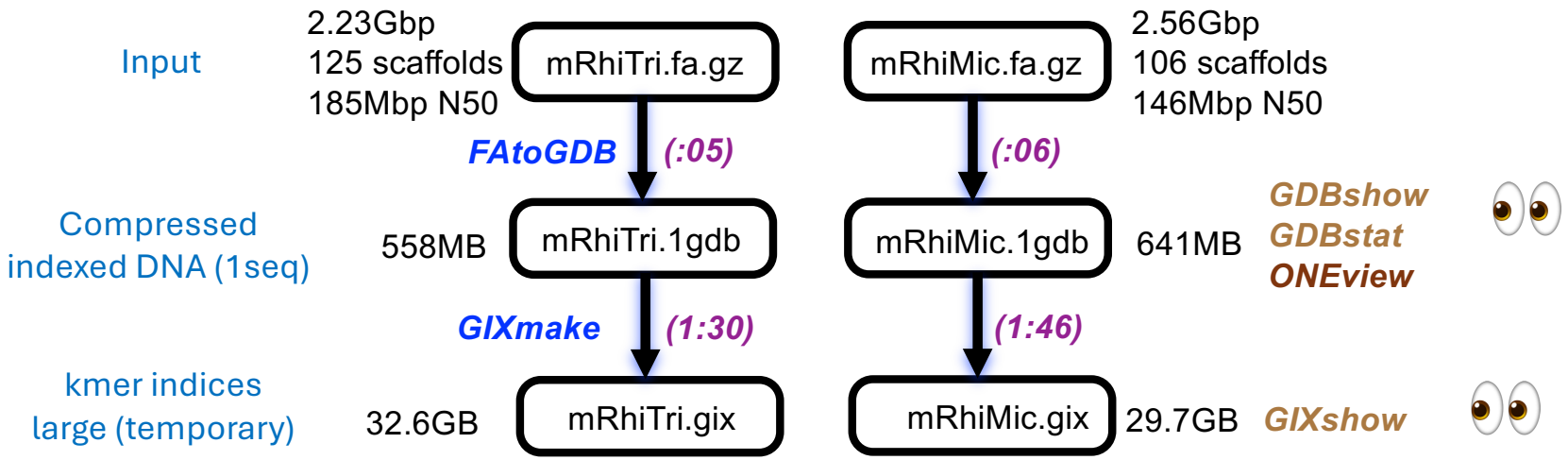
Sort pairs on p_1 (order in G_1)

MSD radix sort is cache local

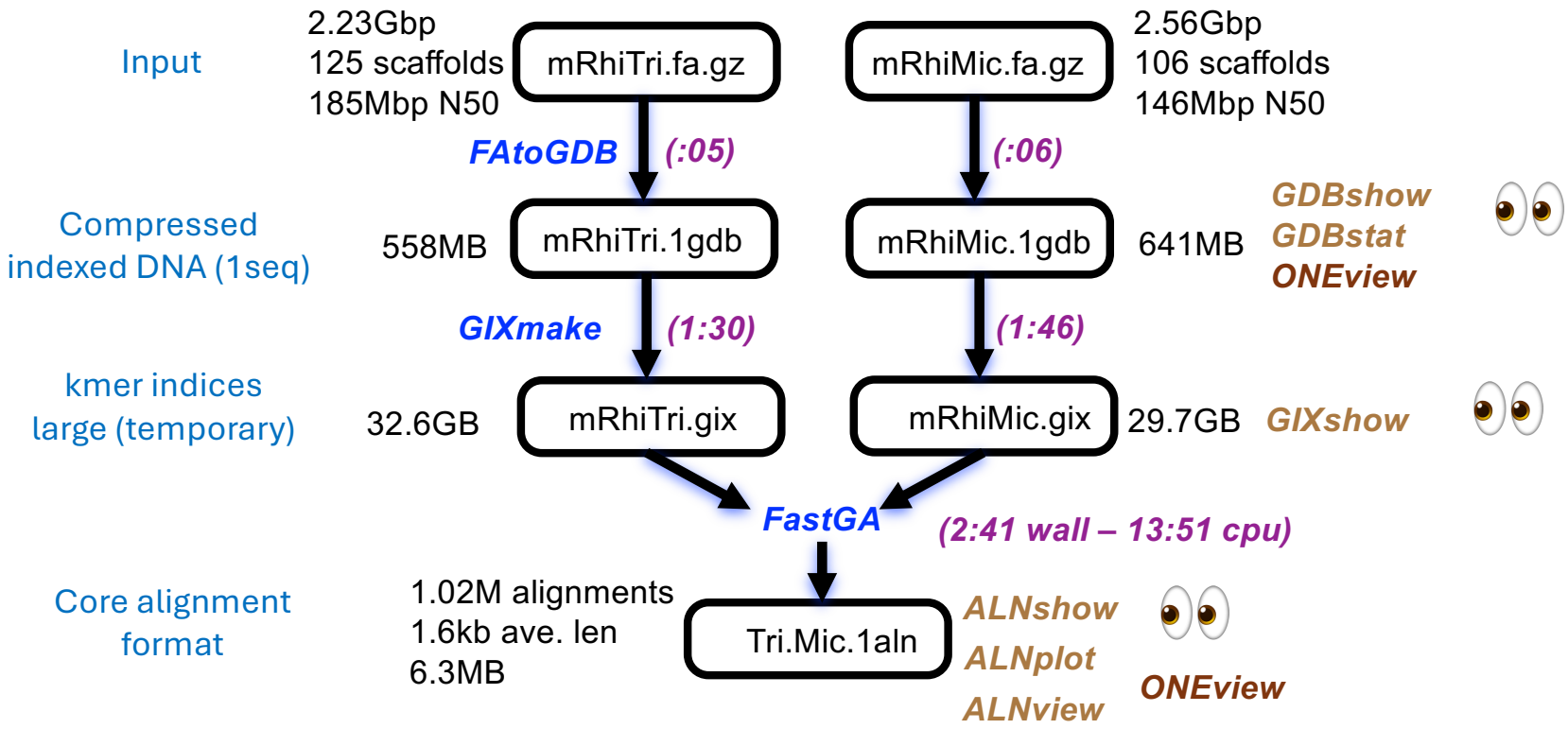
FastGA: data flow, times, disk usage



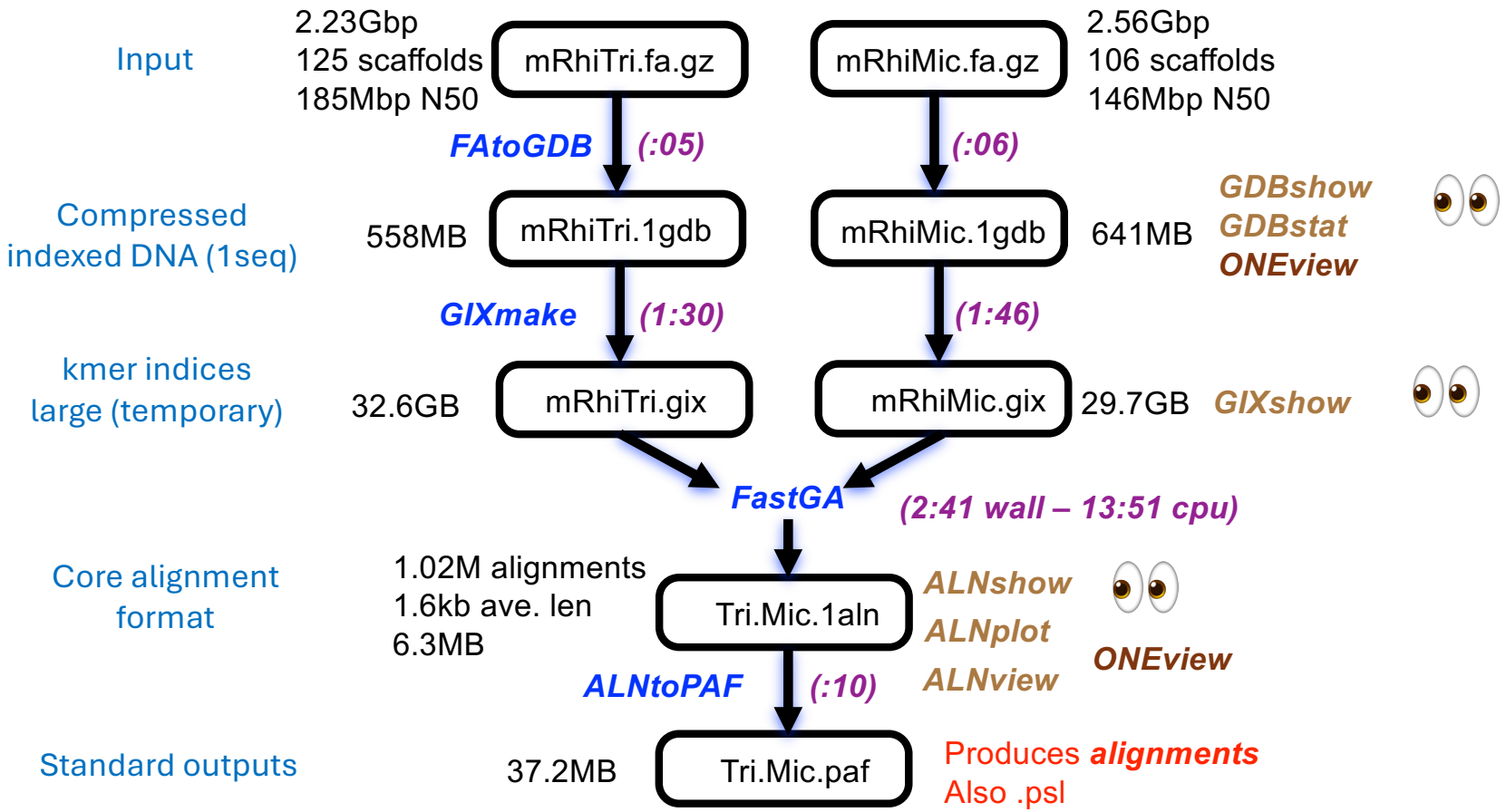
FastGA: data flow, times, disk usage



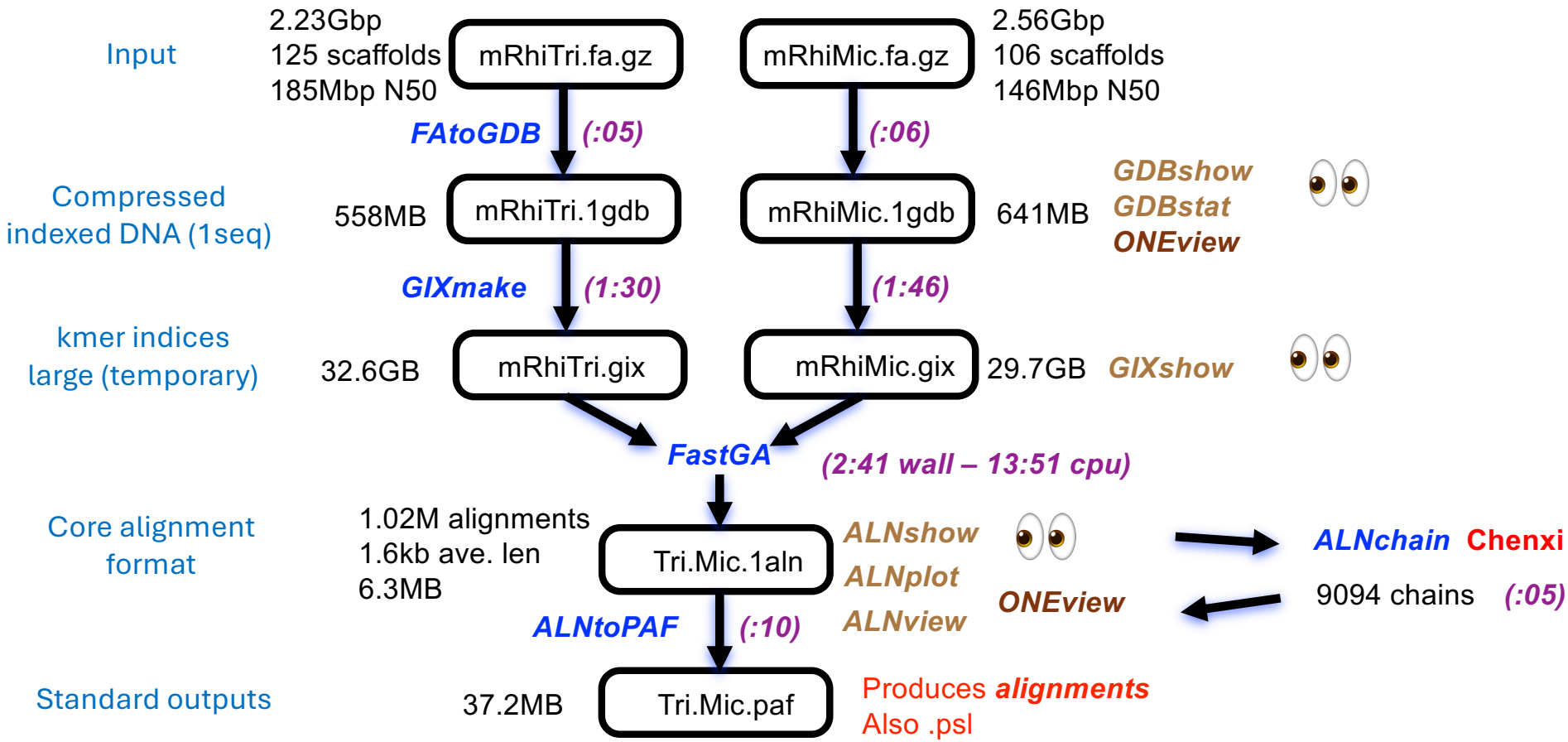
FastGA: data flow, times, disk usage



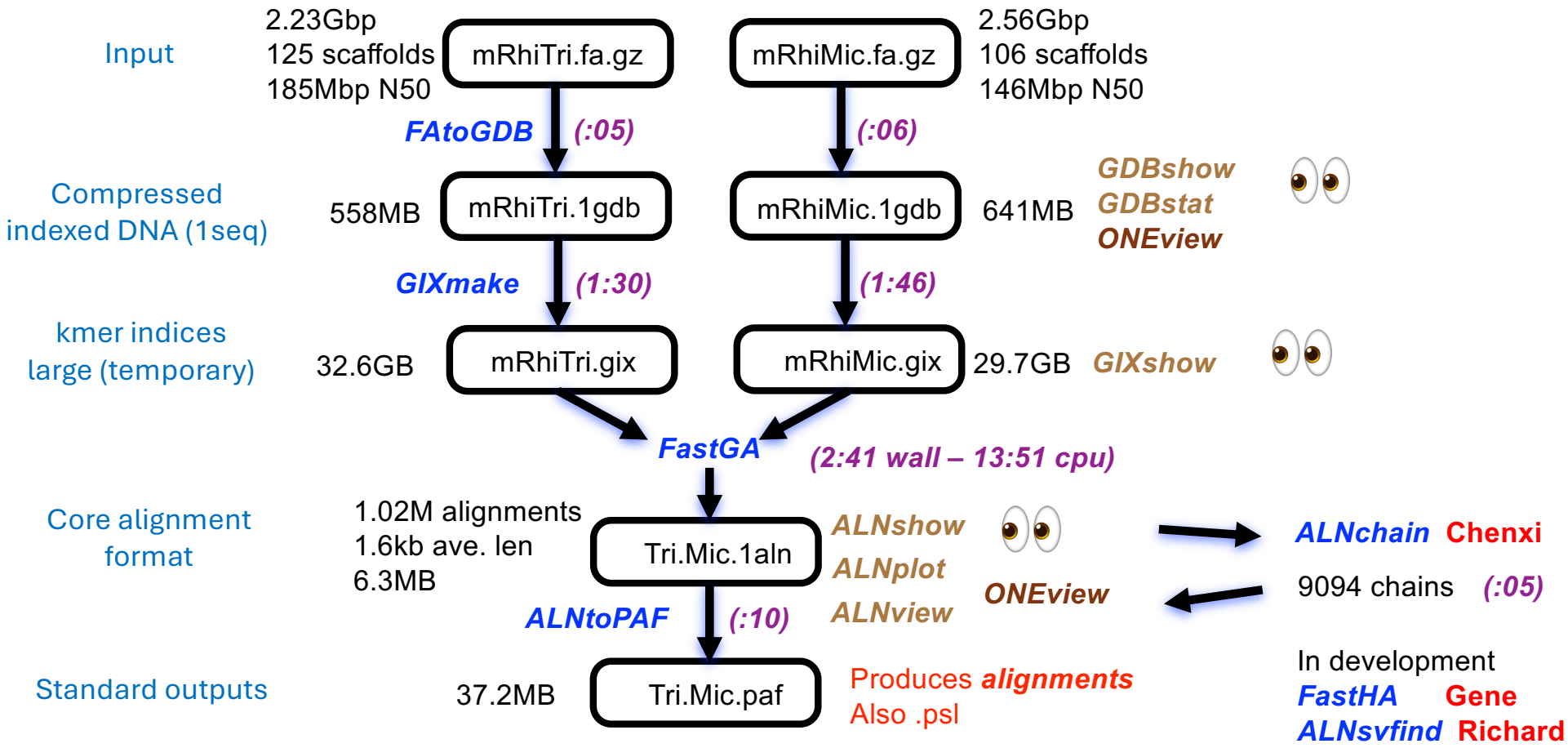
FastGA: data flow, times, disk usage



FastGA: data flow, times, disk usage



FastGA: data flow, times, disk usage



FastGA: Speed and memory comparisons

Between-species comparisons

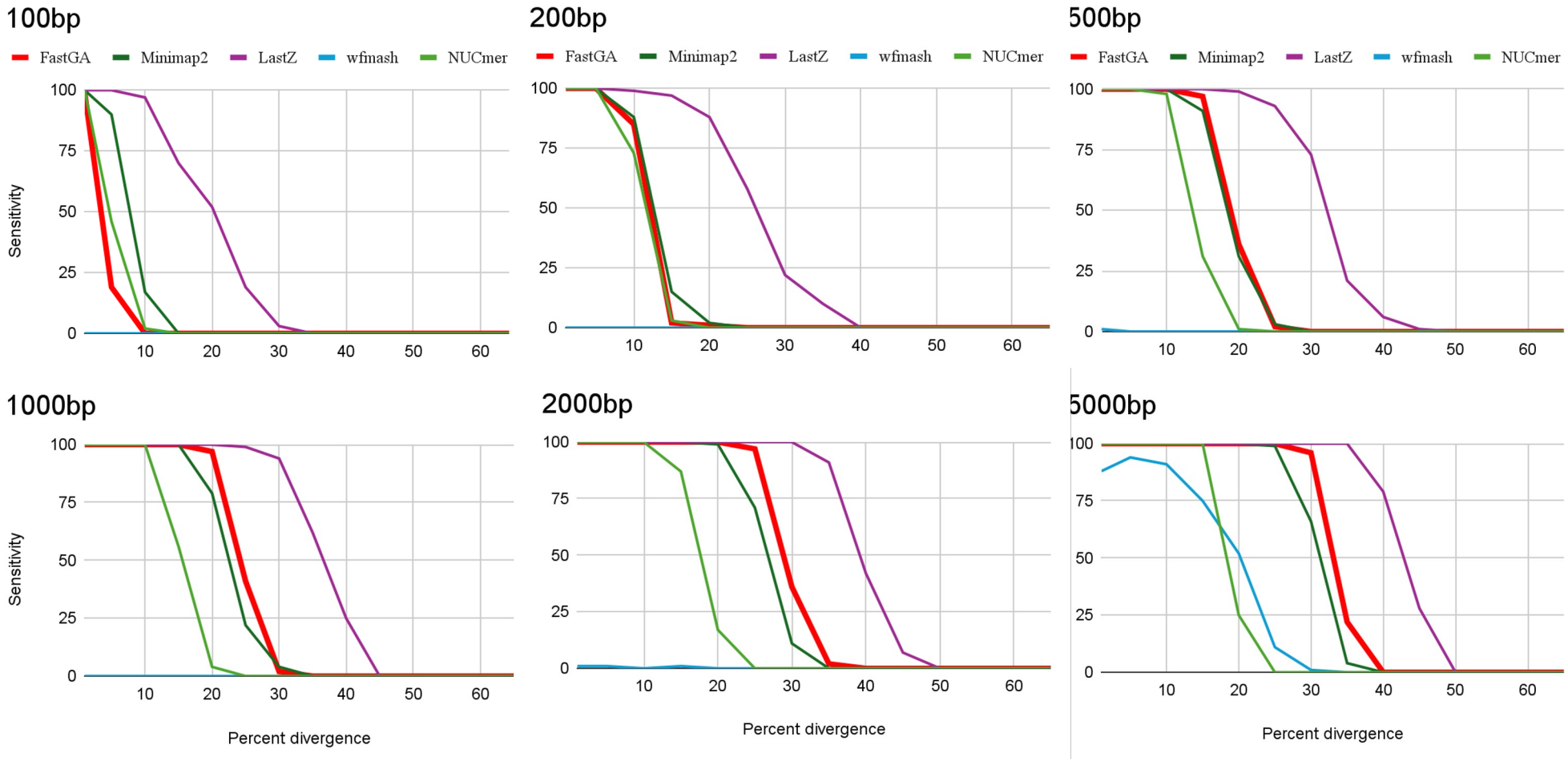
	Assembly name	Size (Mb)	Wall-time (min.)					Peak memory (Gb)				
			Minimap2	Wfmash	Nucmer	LastZ*	FastGA	Minimap2	Wfmash	Nucmer	LastZ*	FastGA
fish	fCheCuc1	656	30	26	25	144	4	32.57	8.83	22.6	0.184	2.007
	fCheLuc1	643										
birds	bAnsEry1	1,279	43	18	42	40	5	21.85	16.63	87.91	0.554	3.624
	bAthNoc1	1,310										
mammals	mGloMel1	2,662	31	44	224	809	8	30.95	32.2	129.48	0.43	6.269
	mLagAlb1	2,547										
newts	aLisHel1	24,147	-	-	-	-	177	-	-	-	-	56.31
	aLisVul1	24,660										

Diploid self-comparison

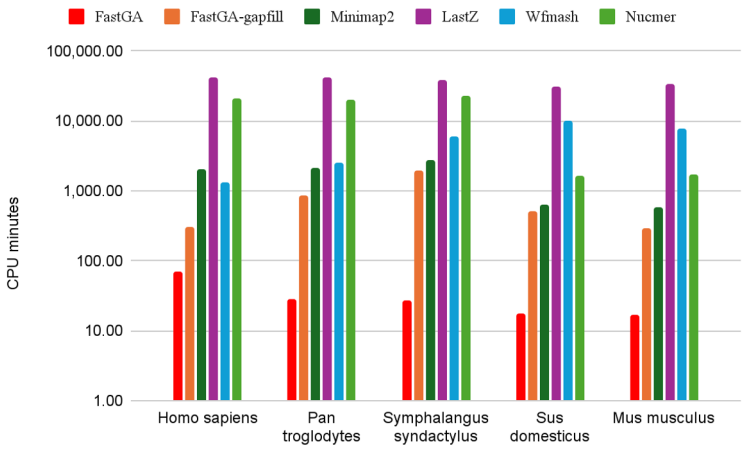
	Assembly name	Size (Mb)	Minimap2	Wfmash	Nucmer	LastZ*	FastGA	Minimap2	Wfmash	Nucmer	LastZ*	FastGA
fish	fCheCuc1	1,319	360	20	-	435	5	35.08	16.22	-	0.181	3.04
bird	bAnsEry1	2,509	899	41	-	963	7	34.66	30.51	-	0.508	5.844
newt	aLisHel1	43,718	-	-	-	-	227	-	-	-	-	105.294

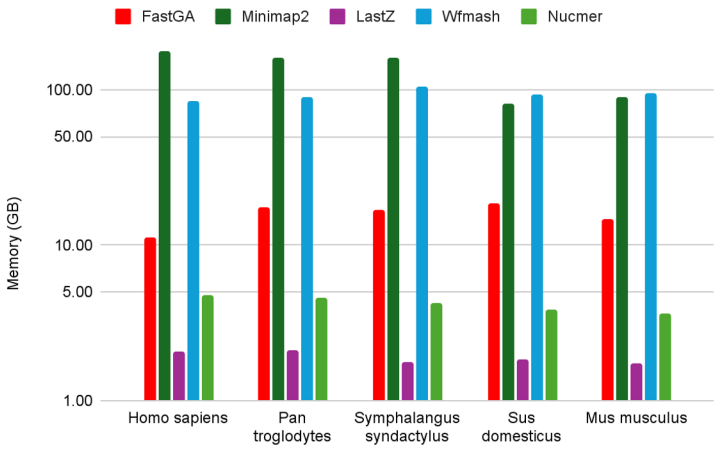
LastZ *: low sensitivity required to complete (--notransition --step=20 --nogapped), only sequences >1Mb

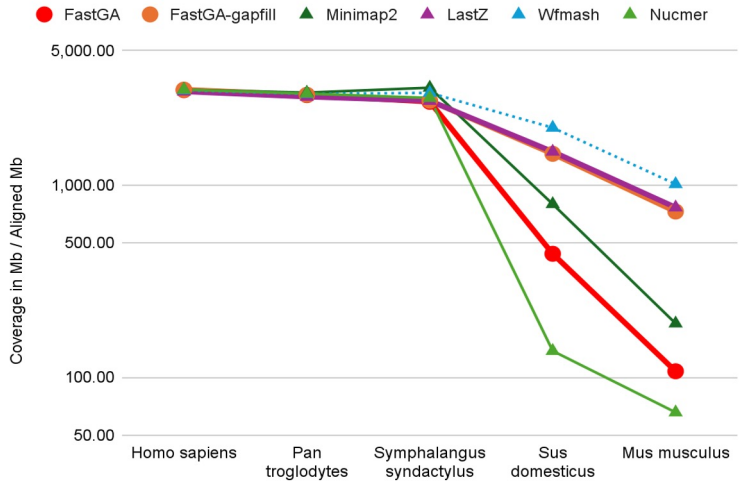
Sensitivity on simulated data



Human genome to other species







Other features of FastGA

- Extension alignment by O(Nd) “wavefront” (Myers, Ukkonen 1986)
  - Levenshtein with linear indels – require >70% match in last 100bp
  - use tracepoints every 100bp to store alignment – very fast and compact
  - minimize gap count in post-processing
- No repeat masking – seed in unique(ish) regions, extend through repeats
  - All standard aligners require to repeat mask first
- Separate aligned-block finding from long range chaining
  - Avoid long affine gaps and alignments through weak (?unalignable) regions

ONEcode (RD with Gene)

- Binary/Ascii
  - Record based, strongly typed, sizes explicit
- Inbuilt type-dependent compression
  - Integers adaptive precision
  - Arrays via Huffman codec
- Index to all object types – nested typing
  - IDs by location 1..N (0 is start of data)
- Single files contain schema, data, indices
- Single C source file – **no** dependencies
  - ONElib.c (4225 lines) / ONElib.h (483 lines)
  - C++, Python, rust interfaces exist

I use it everywhere!

```
[RichardsMacBook2021:~/src/ONEcode/SEQUENCE_UTILITIES] rd% ONEview small.1seq
1 3 seq 2 1
! 4 5 seqio 3 1.0 24 ./seqconvert -1 small.fa 19 2024-08-29_19:14:10
! 4 7 ONEview 3 0.0 18 ONEview small.1seq 19 2024-09-03_15:14:26
.
~ O S 1 3 DNA
~ D I 1 6 STRING
~ D Q 1 6 STRING
~ D N 3 3 INT 4 CHAR 3 INT
.
# S 10
@ S 72
+ S 580
# I 10
@ I 5
+ I 41
# N 2
.
S 51 cttagtagcgatattagttaataaaggtaaattcaaatgcgagtggtagat
I 4 seq1
S 72 ctttaccctccgaggctcttatccaccagaaacttccgccggggtccaggactcttaacgttcttctgtaat
I 4 seq2
S 59 catattctgtcgtaaaatgtagaagaaagtagtagacaactcagaacgatcagaacggt
I 4 seq3
N 13 n 1
S 42 ttttgagcgagagagaatgataagacctcgagggagcttgaa
I 4 seq4
S 55 tttaaatcaaaggccgaagttttttttaagcgacaaagcactttaatatcatatag
I 4 seq5
S 66 agagtgaatatcattaaactagacattcacgatagaaaattagttaattatttctctttccgagta
I 4 seq6
S 49 gctctgtataatgtttctgttttaaactgtgtttgggattatgctaagc
I 4 seq7
N 23 n 2
S 62 ccgagatctataacagtatcaaaaataaaaaactttaataaaatattaaaattattgaaat
I 4 seq8
S 53 tagaagttgtttaataagttttattcacaatcgtttaatatttacacataaaa
I 4 seq9
S 71 acatttacatattgatgtaacactcctatagcctttgatgaccgaaaactttaatttttaatacgagagtt
I 5 seq10
```

Prologue

Schema

Counts/sizes of data line types

Data

https://github.com/thegenemyers/ONEcode

.1aln

```
1 3 aln 2 1
! 4 6 FastGA 3 0.1 59 FastGA -vk -1:Tri-Mic ../mRhiTri/mRhiTri ../mRhiMic/mRhiMic 19 2024-09-03_14:04:12
! 4 7 ONEview 3 0.0 20 ONEview Tri-Mic.1aln 19 2024-09-03_15:52:00
.
< 23 ../mRhiTri/mRhiTri.1gdb 1
< 23 ../mRhiMic/mRhiMic.1gdb 2
< 35 /Users/rd/SEQUENCE-DATA/BATS/FASTGA 3
.
~ D t 1 3 INT          "P 3 aln                        ALIGNMENTS\n"
~ O a 1 3 INT          "D t 1 3 INT                    trace point spacing in a - global\n"
~ G A 0                "O a 1 3 INT                    number of alignments for colinear groups = a chain\n"
~ D p 2 3 INT 3 INT    "G A                            chains (a) group alignment objects (A)\n"
~ O A 6 3 INT 3 INT 3 "D p 2 3 INT 3 INT              spacing in a,b between end of previous alignment and start of next\n"
~ D L 2 3 INT 3 INT    "                               alignment: a_read[beg..end] b_read[beg..end], 0-indexing\n"
~ D R 0                "O A 6 3 INT 3 INT 3 INT 3 INT 3 INT 3 INT\n"
~ D Q 1 3 INT          "D L 2 3 INT 3 INT              lengths of sequences a and b\n"
~ D M 1 3 INT          "D R 0                          flag: reverse-complement sequence b\n"
~ D D 1 3 INT          "D Q 1 3 INT                    quality: alignment confidence in phred units\n"
~ D C 1 6 STRING       "D M 1 3 INT                    match: number of matching bases\n"
~ D T 1 8 INT_LIST     "D D 1 3 INT                    differences: number of diffs = substitions + indels\n"
~ D X 1 8 INT_LIST     "D C 1 6 STRING                 cigar string: encodes precise alignment\n"
                       "D T 1 8 INT_LIST               trace points in b\n"
.                      "D X 1 8 INT_LIST               number of differences in alignment per trace interval\n"
# t 1
# A 1019564
# R 495232
# D 1019564
# T 1019564
@ T 963
+ T 17359026
# X 1019564
@ X 963
+ X 17359026
.
t 100
A 0 1 3748 117 16 3617
D 456
T 38 98 99 102 98 101 99 95 99 95 95 93 99 94 93 92 98 96 91 88 94 94 95 97 94 99 102 98 94 96 98 113 96 93 89 97 95 86 46
X 38 6 8 15 9 10 11 6 6 13 5 7 7 14 11 8 16 6 9 12 6 9 12 15 15 10 10 16 10 14 10 23 18 26 18 12 16 28 9
A 0 6 2994 0 2169 5047
D 284
T 30 93 100 100 96 99 97 100 99 96 95 92 93 92 95 96 98 97 95 90 96 92 95 97 95 99 97 99 98 96 91
X 30 1 0 2 6 8 7 7 5 11 5 8 7 16 9 10 16 5 7 10 4 8 9 15 14 10 11 17 15 19 22
A 0 6 3023 0 2181 5093
D 297
T 31 93 100 100 96 99 97 100 99 96 95 92 93 92 95 96 98 97 95 90 96 92 95 97 95 99 97 99 98 99 100 22
X 31 1 0 2 6 8 7 7 5 11 5 8 7 16 9 10 16 5 7 10 8 8 7 15 14 10 11 17 14 20 27 6
A 0 6 2860 90 9157 11985
R
D 328
T 29 93 100 100 96 99 97 100 99 95 98 92 100 98 98 97 98 97 97 92 97 96 99 101 101 100 112 102 102 72
X 29 1 0 2 6 8 7 7 5 10 7 8 11 18 11 11 16 5 5 8 7 11 13 20 18 11 27 23 31 21
```

Source files – used for sequence ids, lengths, coordinate mapping

A for alignment:

- a, [begin, end)
- b, [begin, end)

D for edit distance

T for trace points

- INT_LISTs are differenced then byte-sized

X for diffs per trace block

R for reverse/complement

.1aln quirks

- The coordinates are currently are **contig** level, where input sequences (scaffolds) are split into contigs by runs of Ns
  - These are what FastGA runs on implicitly, avoiding alignments across contig boundaries
- The mapping from contigs to scaffolds/inputs is governed by information in a **.1gdb** file
  - The **.1gdb** information is also embedded in the top of the **.1aln** files, so that tools have access for coordinate transformation without external reference

Contents of .1gbwt file (~GFA): header

```
rd@farm22-head2:~/DATA/HPRC/release1-fa$ ONEview -H HPRC-63.1gbwt
1 3 seq 2 1
2 4 gbwt
! 4 4 syng 3 1.0 4319 syng -readK HPRC.1khash -zeroK -o HPRC-4 -writeGBWT -writeEnds HG00438…
< 45 HG00438.maternal.f1_assembly_v2_genbank.fa.gz 1
…
~ D h 3 3 INT 3 INT 3 INT   k, w, seed for the seqhash: for syncs k = |smer|, w+k = |syncmer|
~ O S 1 3 DNA               sequence of the node
~ O V 1 3 INT               graph node (vertex): length
~ D K 1 3 INT               coverage count of node
~ D E 4 4 CHAR 3 INT 3 INT 3 INT   edge: +/-, adjacent node (- if reversed), offset, count
~ D B 2 4 CHAR 8 INT_LIST          GBWT: +/-, list of node indices from opposite-signed E
lines
~ D C 2 4 CHAR 8 INT_LIST          GBWT: +/-, list of run-length counts
~ O P 2 3 INT 3 INT         path: reference file number and sequence number in file
~ D Z 2 3 INT 3 INT         GBWT path: starting node, starting count, length in nodes
~ D z 1 8 INT_LIST          alternative explicit list of node ids (-ve if reversed)
~ D d 1 6 STRING            directions (orientations) of nodes in path (optional): 1:1 with P
~ D o 1 8 INT_LIST          offsets of the nodes from start of sequence, 1:1 with P
~ D X 1 3 DNA               prefix before first node - required to fully reconstruct
~ D Y 1 3 DNA               suffix after last node - required to fully reconstruct
.
# V 192600025        number of nodes
…
# E 457929032        number of edges (counted twice, once for each end)
…
# B 46059745         number of run-length encoded GBWT lines
…
# P 37269            number of paths (encoded sequences)
```

Complete count info

```
# V 192600025
% V # B 2
% V + B 287029
% V # C 2
% V + C 287029
% V # E 13888
# E 457929032
# B 46059745
@ B 189324
+ B 323129517
# C 46059745
@ C 189324
+ C 323129517
# P 37269
% P # X 1
% P + X 54
% P # Y 1
% P + Y 55
% P # Z 1
# Z 37269
# X 37269
@ X 54
+ X 732424
# Y 37269
@ Y 55
+ Y 766419
```

count
@ max
+ total sum

Contents of .1gbwt file (~GFA): data (here in ascii, default is indexed binary)

```
rd@farm22-head2:~/DATA/HPRC/release1-fa$ ONEview -h HPRC-63.1gbwt | less
…
P 3 218
X 18 acataaaatatcaaagta
Y 38 tgcggtacataacatatcaaagtacccatactatatag
Z -125259621 0
P 3 219
X 0
Y 33 aatcgtcatagaatggaatcgaatggattcatt
Z 29552691 13
…
…
V 63
E + 41 15 1
E - 39 18 1
V 63
E + -42 2 1
E - 40 15 1
…
V 63
E + -48 55 69
E + -130642443 55 2
E + -165916110 55 1
E + -180265576 55 1
E + -180499460 55 2
B + 3 0 1 0
C + 3 1 1 73
E - 50 43 74
E - -120653510 43 1
B - 11 0 1 0 4 0 3 0 1 0 2 0
C - 11 8 1 25 2 24 1 2 1 4 1 6
…
```

paths 218 and 219 from file 3

X and Y are starting/ ending free sequences
Z is starting node (-ve for revcomp), 13 is start count

a couple of simple nodes (V for vertex) indexed implicitly by position in file, 63 is node length

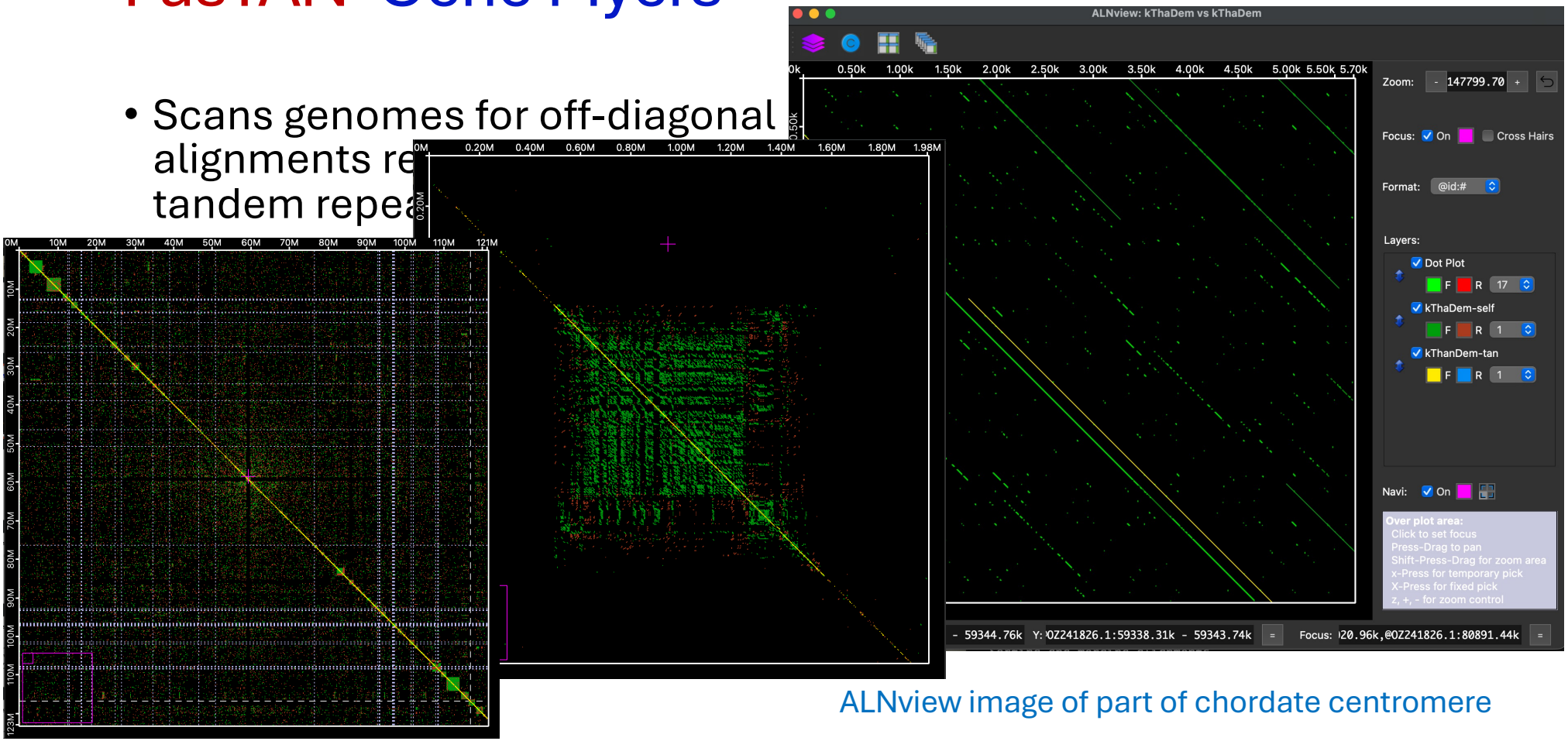
E are edges: +/- for leave/join, node index, offset, count

a complex vertex, with multiple incoming and outgoing edges

B lines are lists of edge indexes, lengths 3 and 11
C lines are corresponding run length counts

FasTAN Gene Myers

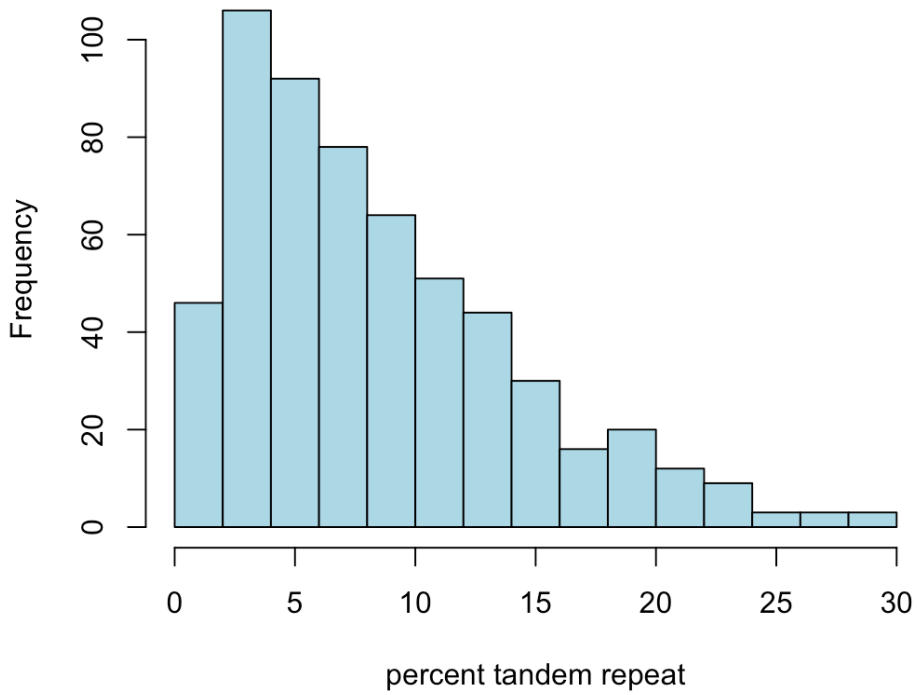
- Scans genomes for off-diagonal alignments re
  tandem repea



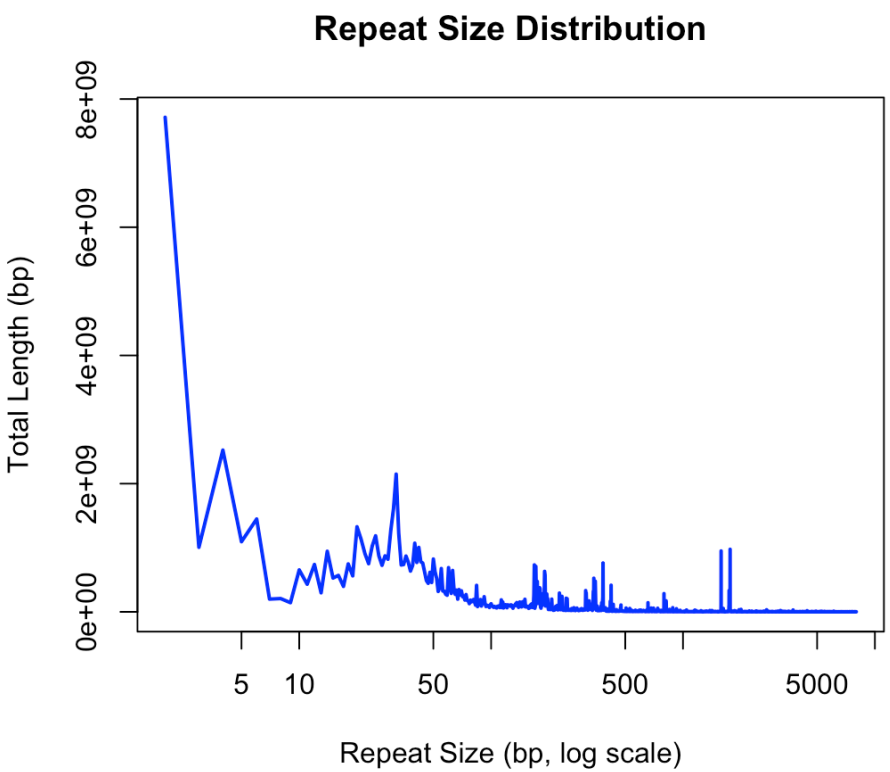
ALNview image of part of chordate centromere

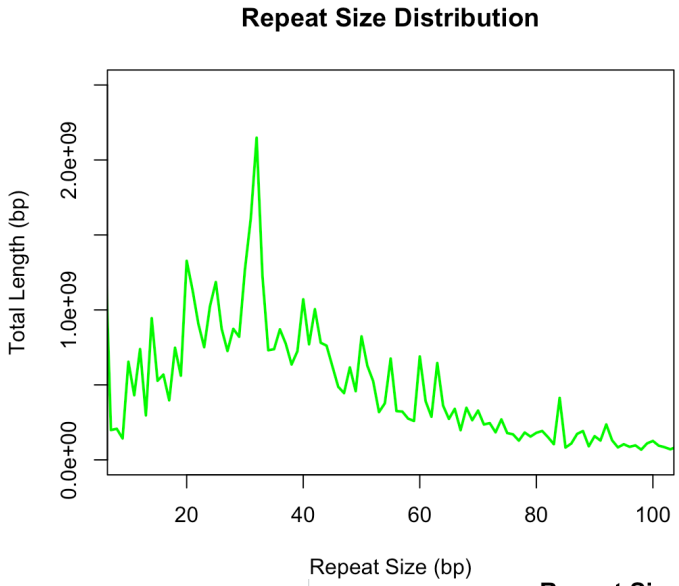
FasTAN run on 577 VGP assemblies

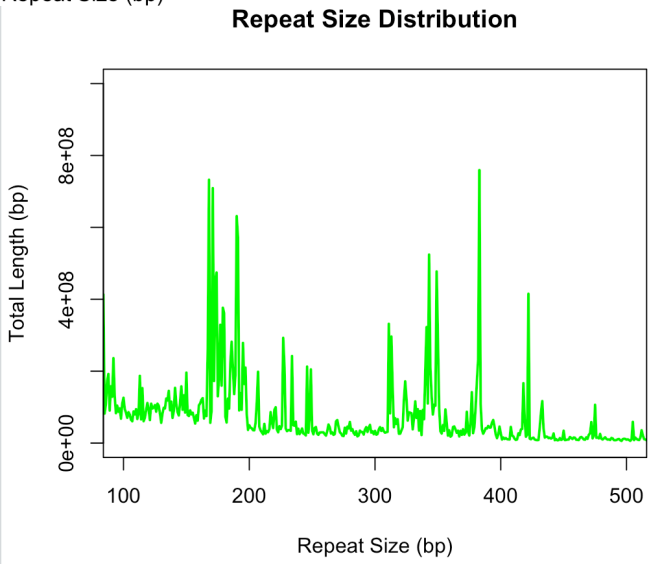
- Large variation in total amount of tandem repeats
  - Min 0.8 %
  - Max 29.5%
  - Mean 8.5%



Distibution repeat size





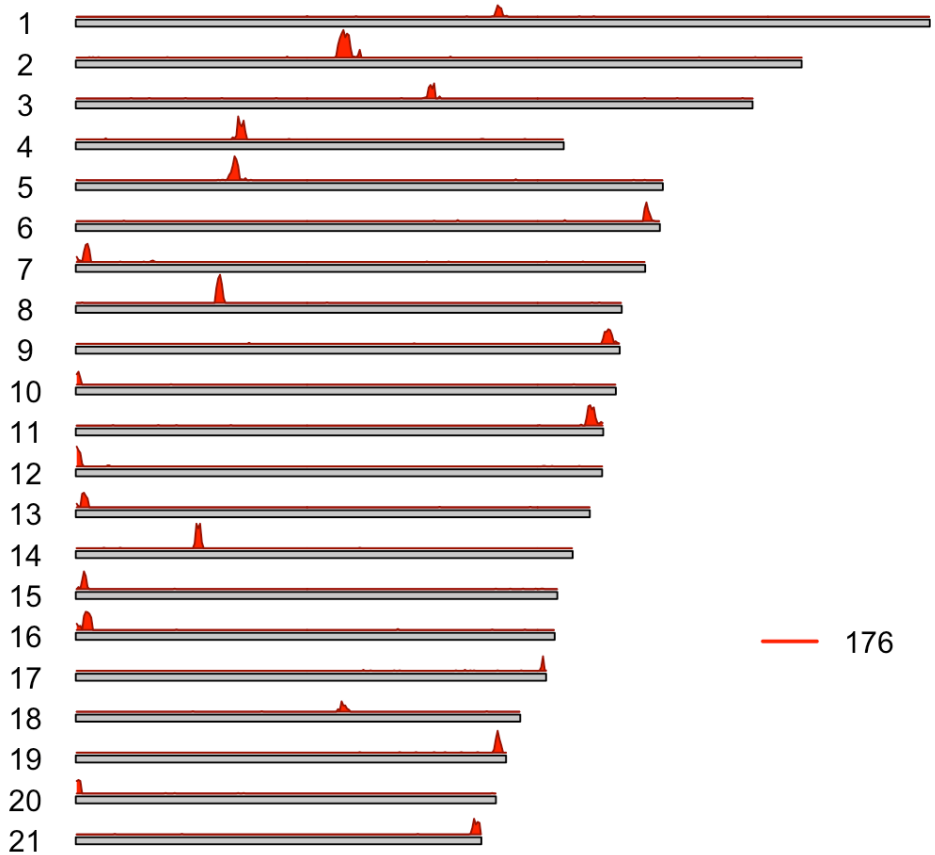


fSolSol = *Solea solea* = common sole



- Look for repeat unit sizes that (tend to) have one tight peak per chromosome
  - Low spatial variance per chromosome, on many chromosomes
  - Have high mean local density (in 100kb windows)
  - Cover at least a few Mb

name	unit	count	bp	dens.
fishes/Solea_solea	39	408	281467	0.21
fishes/Solea_solea	178	40	171626	0.35
fishes/Solea_solea	38	4993	11646162	0.74
fishes/Solea_solea	179	36	100044	0.24
fishes/Solea_solea	177	967	661044	0.17
fishes/Solea_solea	176	1662	7156734	0.68
fishes/Solea_solea	141	2350	2013059	0.28



Summary Pipeline

Initial tandem arrays

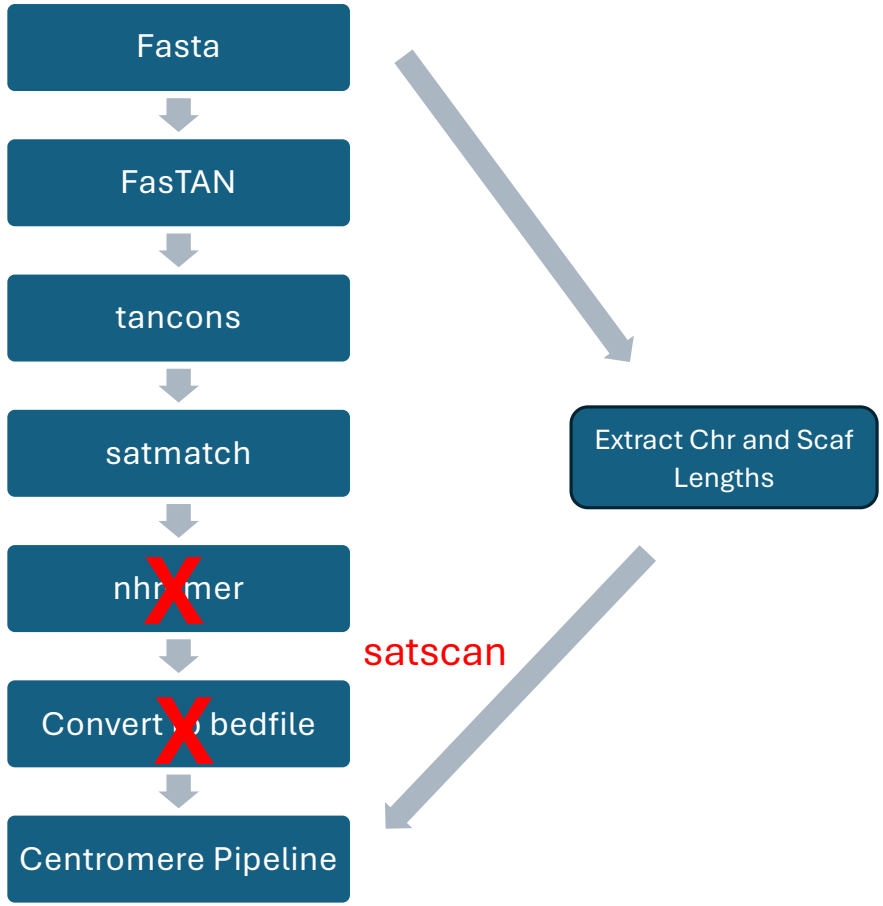
Consensus per array

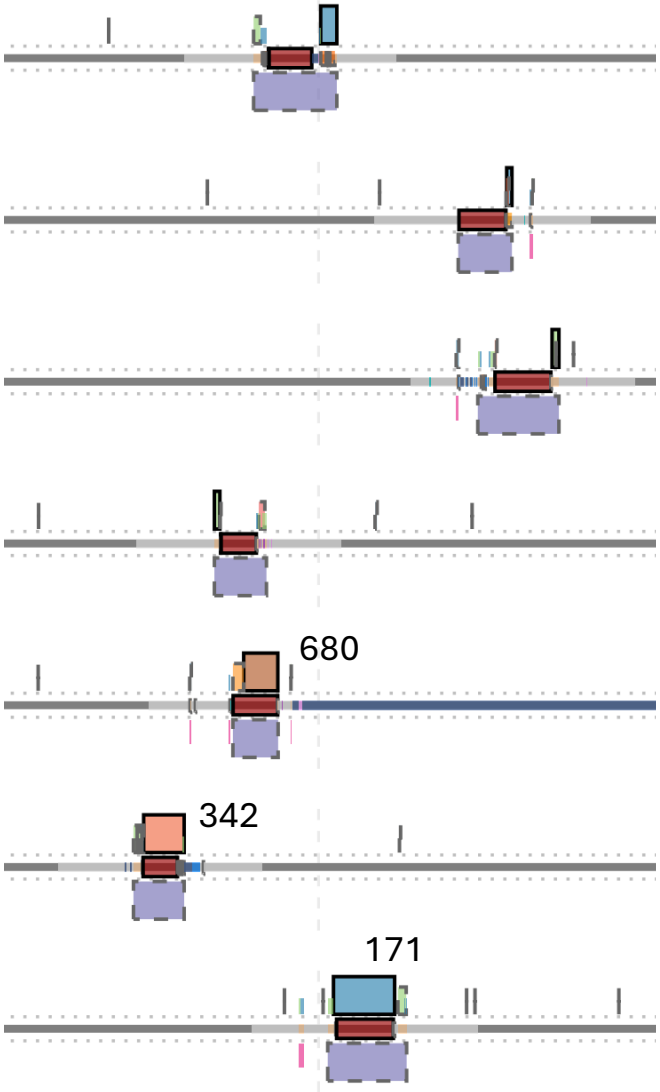
Cluster and consensus

Sensitive alignment

Bed file

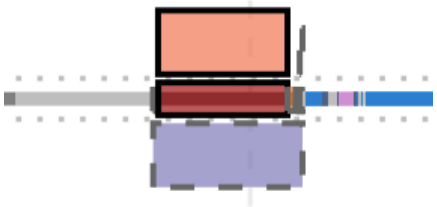
Centromere candidates





Pipeline successfully identifies centromere

FasTAN alone provides incomplete picture



FasTAN (343bp)

Ground truth (CHM13 website)

nHmmer consensus
built from FasTAN hits

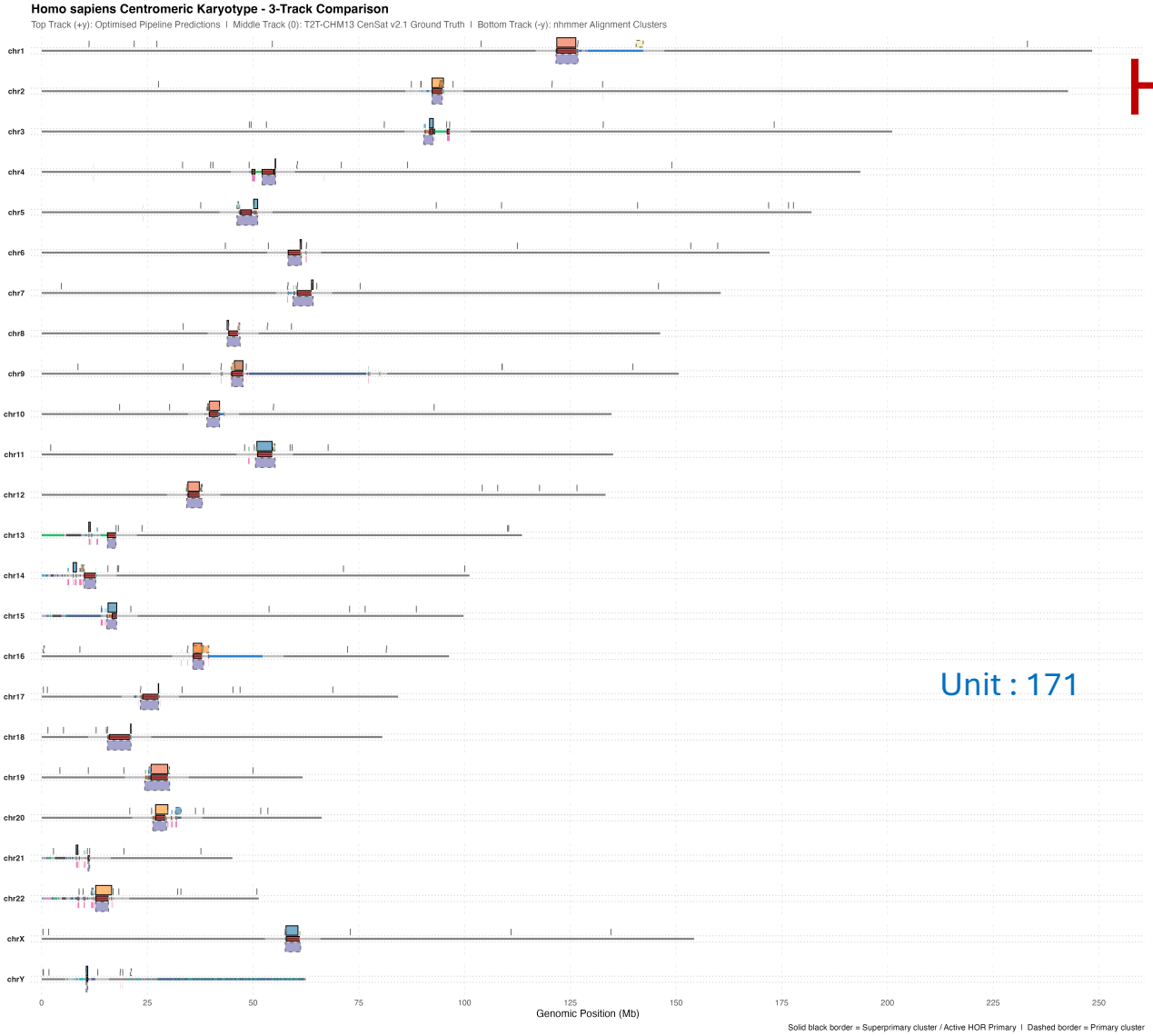
- Pipeline: 125 bp
- Pipeline: 171 bp
- Pipeline: 172 bp
- Pipeline: 188 bp
- Pipeline: 217 bp
- Pipeline: 340 bp
- Pipeline: 342 bp
- Pipeline: 343 bp
- Pipeline: 433 bp
- Pipeline: 468 bp
- Pipeline: 680 bp

- CenSat: Active HOR (Primary)
- CenSat: Active HOR (Secondary)
- CenSat: Diverged HOR
- CenSat: Monomeric Alpha
- CenSat: HSat1A
- CenSat: HSat1B
- CenSat: HSat2
- CenSat: HSat3
- CenSat: BSat
- CenSat: GSat
- CenSat: CT (Transition)
- CenSat: Other CenSat
- CenSat: rDNA

- nhmmer: Primary Cluster (171 bp)
- nhmmer: Secondary Cluster (171 bp)

Cluster Type

- Superprimary
- Primary
- Secondary



Homo sapiens

Unit : 171

Feature & Annotation Category

- Pipeline: 125 bp
- Pipeline: 171 bp
- Pipeline: 172 bp
- Pipeline: 188 bp
- Pipeline: 217 bp
- Pipeline: 340 bp
- Pipeline: 342 bp
- Pipeline: 343 bp
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- CenSat: HSat3
- CenSat: BSat
- CenSat: GSat
- CenSat: CT (Transition)
- CenSat: Other CenSat
- CenSat: rDNA
- nhmmer: Primary Cluster (171 bp)
- nhmmer: Secondary Cluster (171 bp)

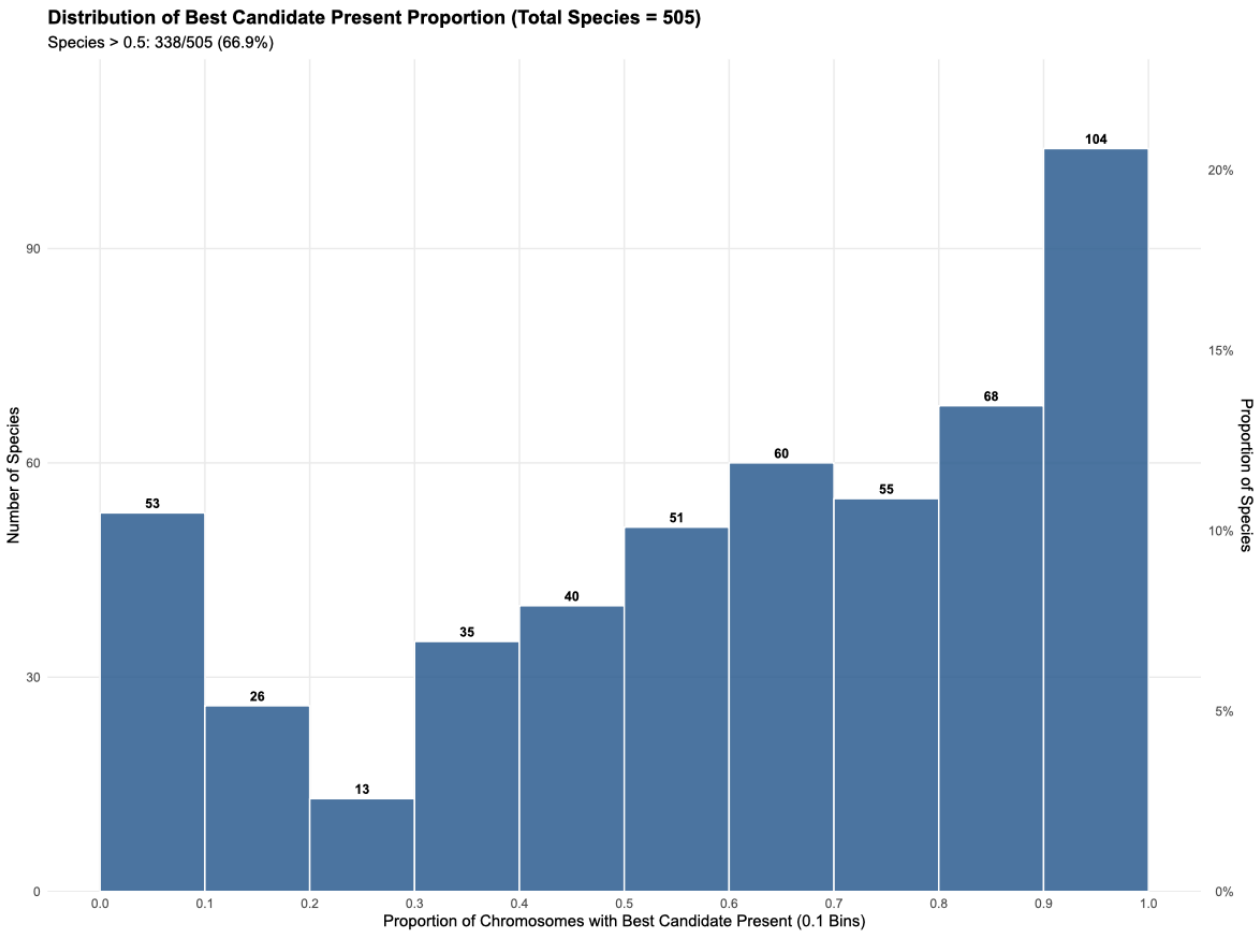
Cluster Type

- Superprimary
- Primary
- Secondary



For what fraction of species/chromosomes do we have candidates?

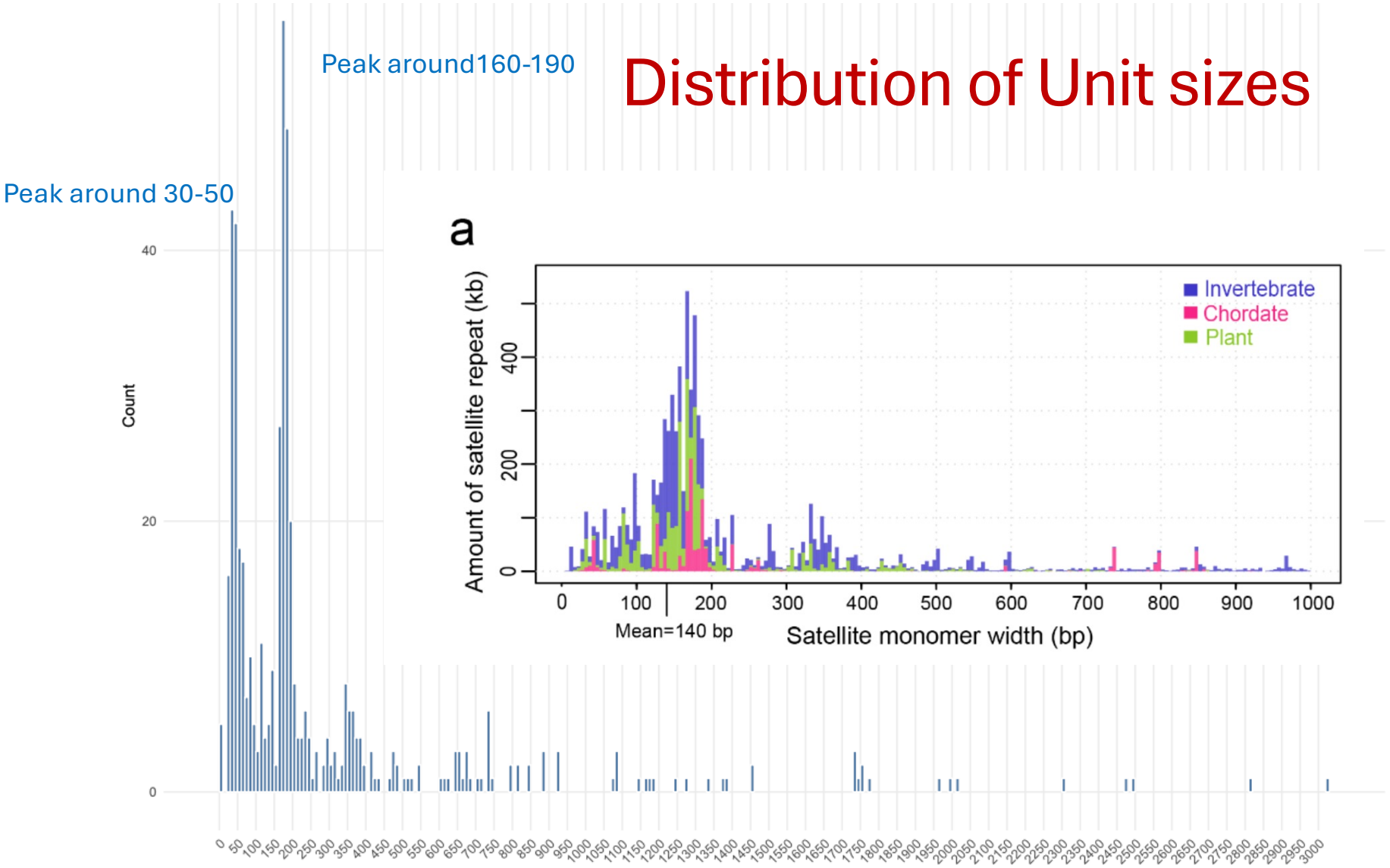
- We find candidates for a varying fraction of the chromosomes per species.
- For 67% of species, we find the best candidates present on over 50% chromosomes
- Overall, the best candidate is present on: 7973/13743 = 58.01% of chromosomes



Distribution of Unit sizes

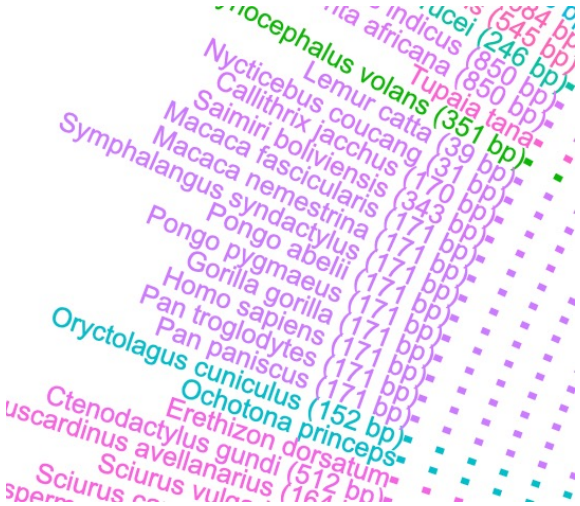
Peak around160-190

Peak around 30-50



All species

Primates: 171, 343





Groups have similar repeat unit sizes

Wall Lizards: 185/186



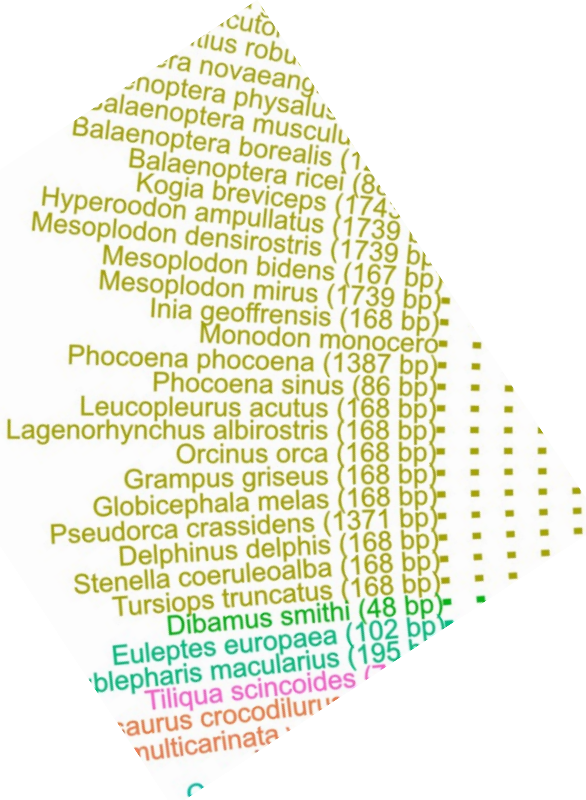
Perciformes: 186/176



Cranes: 190/191



Cetacea: 168/1739



Practical

1. Run FastGA on two Arabidopsis genomes – col0 and cvi0
   - FastGA -v col0 cvi0 > col0-cvi0.paf
   - Explore the resulting file
   - FastGA –vk -1:col0-cvi0 col0 cvi0
   - Explore the resulting **files**
   - (if on a Mac) run ALNview
2. Run FasTAN on col0
   - Then tanbed, tancons, satmatch

FastGA on Arabidopsis